

Research Methods:

These slides have been adapted by those originally prepared by
Dr. Susan Hall at the University of Delaware

Developing the Methods Section

This section of both the proposal and final paper addresses how you plan to attack the problem (purpose of the study) and how you will collect data.

Proposal: written in future tense

Paper: written in past tense

Developing the Methods Section

Contents: (TITLED SUBSECTIONS!)

Participants

Procedures

Statistical Analysis

Factors to consider in planning the methods for a study:

- Universality - the methods should be such that they could be carried out by another competent researcher
- Replicability - the results should be the same if the study is repeated on a similar group of participants

Factors to consider in planning the methods for a study:

- Control - any change in the dependent variable should be solely due to the effects of the independent variable;
(lack of control poses a threat to internal validity!)
- Measurement - should be able to quantify the dependent variables, (except in qualitative research)

Factors to consider (and specify in writing) in planning the participants for a study:

- How many participants are needed?
- How many are available?
- Statistical power is increased with increased sample size (power relates to the ability to detect a difference)
- Cost
- Time

Factors to consider (and specify in writing) in planning the participants for a study:

- What are the important characteristics of the subjects
- How are you defining the subject population?
- What screening criteria will you use?

Factors to consider (and specify in writing) in planning the participants for a study:

- How will participants be selected?
- Random selection
(everyone in the population has an equal chance of being selected)
 - insures a true representation of the population
 - promotes the generalizability of the results

Factors to consider (and specify in writing) in planning the participants for a study:

- How will participants be selected?
- Use of volunteers
 - may be a limitation for the study due to the characteristics or motives of the participants
 - may constitute a threat to internal validity

Factors to consider (and specify in writing) in planning the participants for a study:

- How will participants be assigned to groups?
- Use of volunteers
 - random assignment to groups should be used if possible
 - lack of random assignment may constitute a threat to internal validity

Factors to consider (and specify in writing) in planning the participants for a study:

- Informed consent should be obtained from all participants (a sample Informed Consent Form should be included in an appendix of your thesis/dissertation; all informed consent forms should be retained for a period of at least three years)

Factors to consider (and specify in writing) in planning the apparatus and/or instruments for a study:

- All apparatus should be described in detail (including manufacturer name and address, model number, etc. - what about if you made it? Ideally, publish this and then cite)
- Validity and reliability values for all published instruments should be stated

Factors to consider (and specify in writing) in planning the apparatus and/or instruments for a study:

- All instruments newly developed for a study should be:
 - modeled after similar instruments used in previously published research, if possible
 - reviewed by a panel of experts
 - pilot tested on hypothetical participants

Factors to consider (and specify in writing) in planning the procedures for a study:

- Who will collect the data? (researcher, trained personnel, computer? You will likely have a interdisciplinary team!)
- the Experimental Protocol (description of the series of events for each participant throughout the course of the study, described in systematic detail)

Factors to consider (and specify in writing) in planning the procedures for a study:

- the Experimental Protocol
- If a standard protocol is used, must reference the source
- A copy of any standardized instructions read to participants should go in an appendix
- It is important to show that systematic procedures are to be employed

Factors to consider (and specify in writing) in planning the procedures for a study:

- Pilot studies (limited preliminary studies conducted with small samples of participants; may be referred to in one or more subsections of the Methods section)

Factors to consider (and specify in writing) in planning the procedures for a study:

- Pilot studies - conducted for one or more of the following reasons
 - to try out or practice procedures or protocol
 - to try out or calibrate apparatus or instruments
 - to acquire new insights with regard to the design of the study
 - feasibility!